Customer Shopping Behavior Analysis

End-to-end data analysis using SQL, Python, and Power BI

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Executive Summary

3,900	₹1.95Cr	₹4,990	43%
Total Customers	Total Revenue	Avg Purchase	Discount Rate
18 attributes analyzed	₹1,94,62,264	Per transaction	Orders with discount

Complete pipeline: Python → PostgreSQL → Power BI

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Python Data Preprocessing

Key Steps

- Load and inspect dataset
- Handle 37 missing review ratings
- Impute using category median
- Feature engineering
- Push to PostgreSQL

Features Created

- Age groups (quartiles)
- Purchase frequency (days)
- Drop redundant promo codes

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SQL Analysis Highlights

Gender Revenue Split

Male: ₹1.32 Cr (67.7%)

Female: ₹62.8 L (32.3%)

Top Products

Gloves: 3.86 avg rating

Sandals: 3.84 avg rating

Shipping Impact

Express: ₹5,050

Standard: ₹4,881

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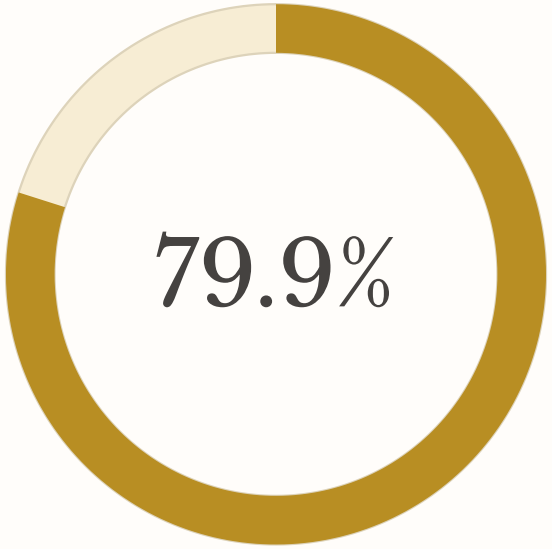
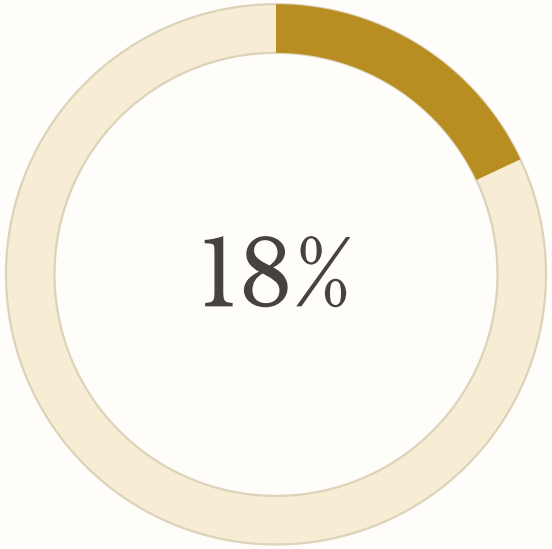
Customer Segmentation



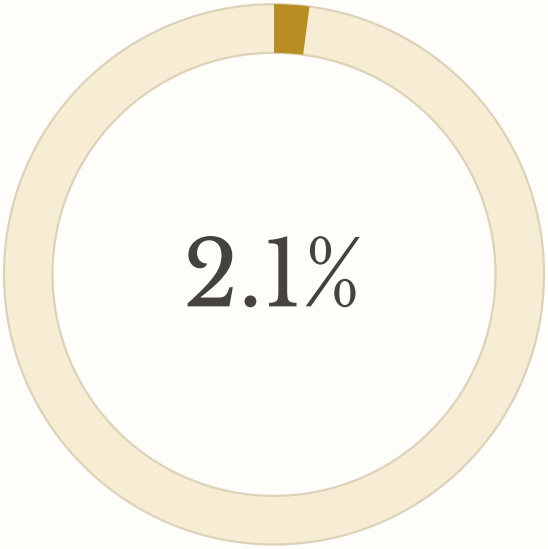
Loyal Customers

11+ previous purchases



Returning

2–10 previous purchases



New Customers

First purchase only

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Subscription Program Analysis

Key Finding

Subscribers spend **₹4,967** vs non-subscribers **₹4,999**

Program needs redesign with minimum spend thresholds or tiered rewards

Repeat Buyers

72.5% of loyal customers (>5 purchases) never subscribed

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Revenue by Category

44.7%

Clothing

₹87,06,044

31.8%

Accessories

₹61,95,700

15.5%

Footwear

₹30,13,766

7.9%

Outerwear

₹15,46,754

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Revenue by Age Group

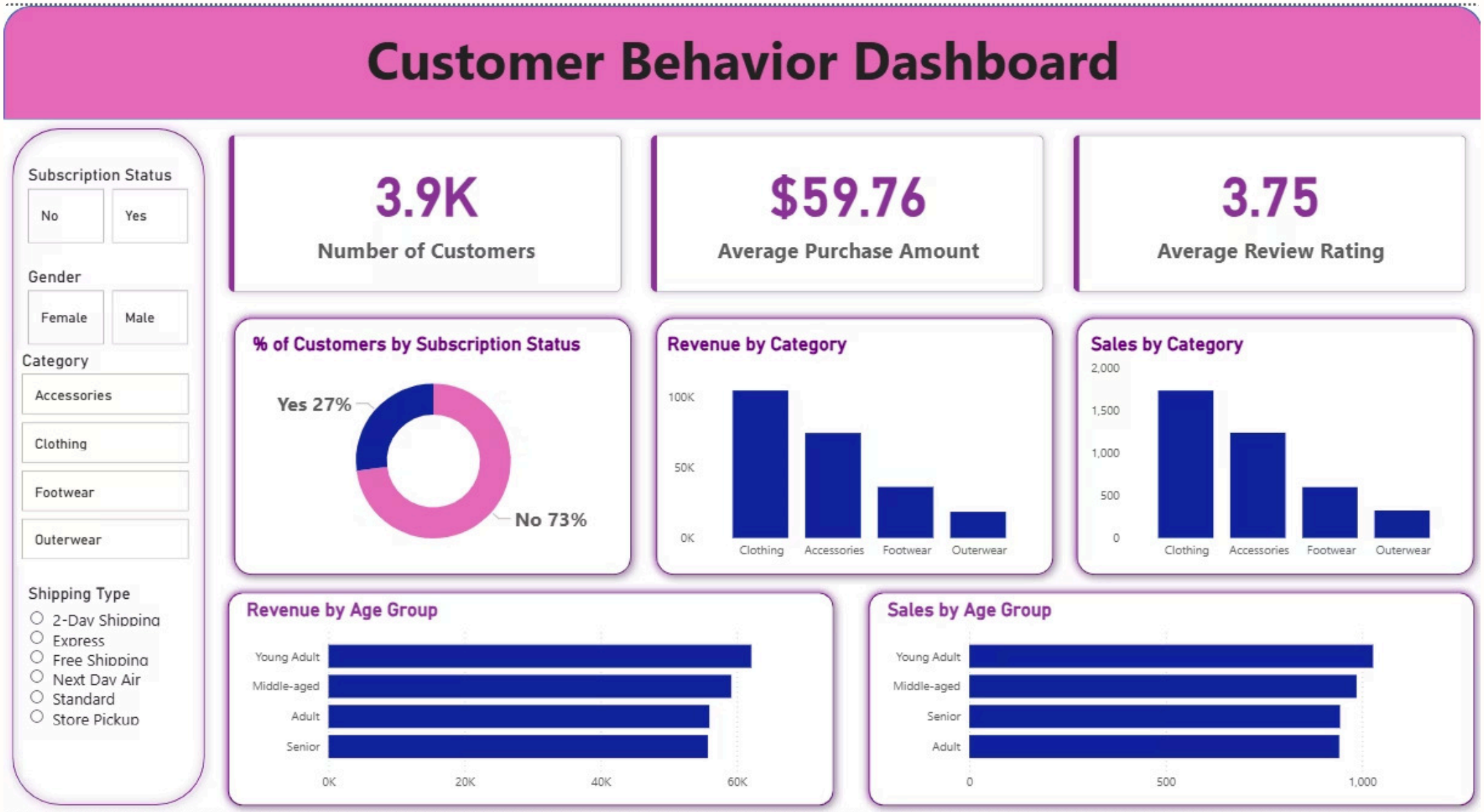
Young Adult (18-31)
₹51,88,940 • 26.7%

Middle-aged (44-57)
₹49,42,950 • 25.4%

Adult (31-44)
₹46,74,163 • 24.0%

Senior (57-70)
₹46,56,210 • 23.9%

Young Adults generate highest revenue — invest in youth-focused products



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Key Findings & Recommendations

01

Launch female-targeted campaigns

Males generate 2.1× more revenue

02

Shift to flash sales

Hat and sneakers discounted 50% of orders

03

Invest in acquisition

Only 2.1% new customers

04

Target repeat buyers

72.5% of loyal customers not subscribed

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Tools & Methodology

Python/Pandas

Data cleaning & feature engineering

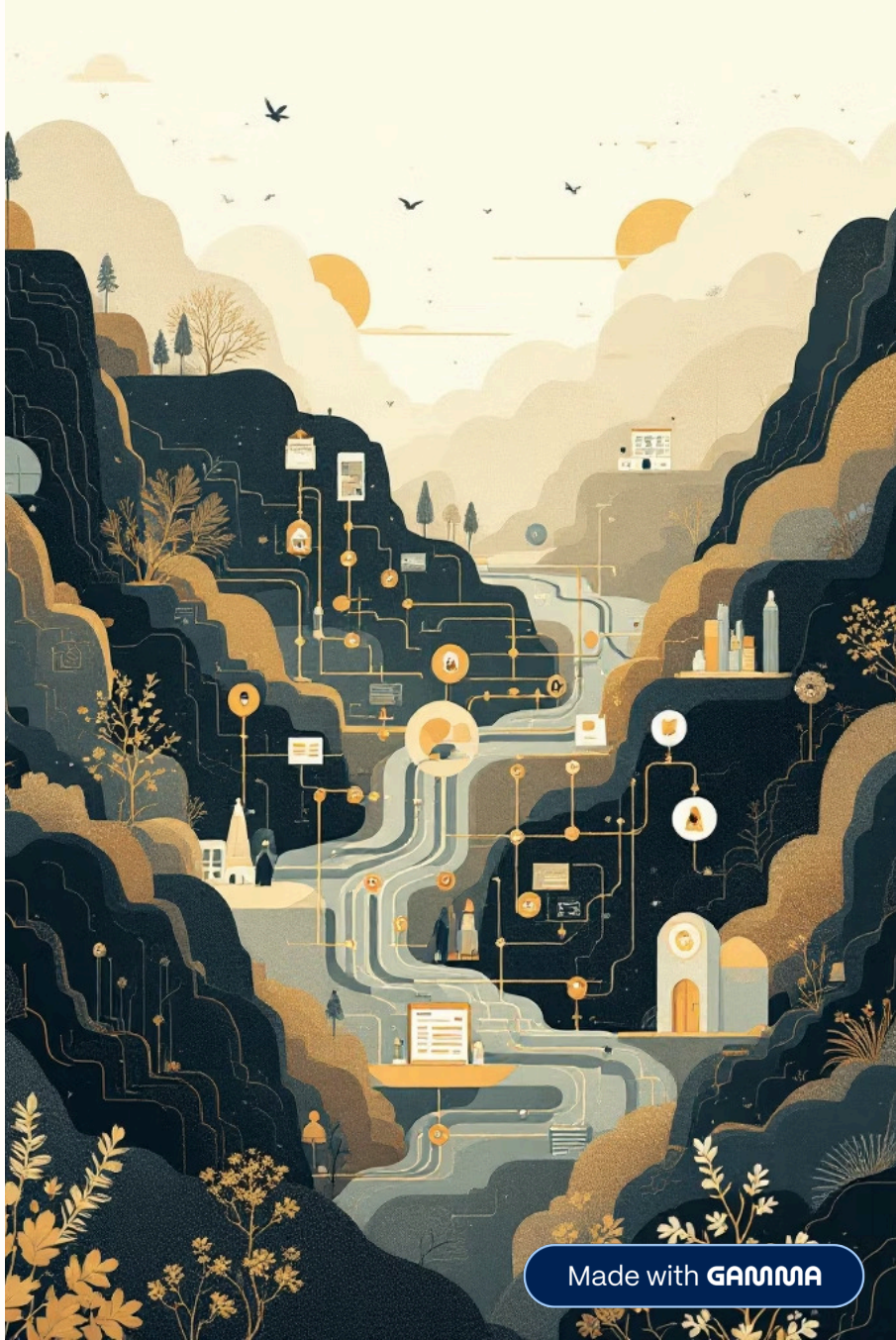
PostgreSQL

SQL analysis with 10 business questions

Power BI

Interactive dashboard with 8 visuals

Complete pipeline mirrors industry-standard data analyst workflows



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